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An optical remote sensing approach for ecological monitoring of red and green *Noctiluca scintillans*

S. K. Baliarsingh · R. M. Dwivedi · Aneesh A. Lotliker · K. C. Sahu · T. Srinivasa Kumar · S. S. C. Shenoi

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Abstract An ecosystem disruptive bloom of red *Noctiluca scintillans* (hereafter *Noctiluca*) was observed in coastal waters of the north-western Bay of Bengal during April 2014. Based on the principle of phytoplankton group/species specific remote sensing reflectance (R_{rs}), a technique of detecting green *Noctiluca* and diatom was developed earlier using R_{rs} at 443, 488, and 531 nm of Moderate Imaging Spectroradiometer-Aqua (MODIS). This was appropriately modified to detect bloom of red *Noctiluca* in coastal waters of the Bay of Bengal. Additional R_{rs} data at longer wavelengths viz. 667 and 678 nm were included in the existing algorithm, and the spectral shapes were accounted to detect the bloom of red *Noctiluca*. The classification scheme discriminates red *Noctiluca* from the green form of the same species and diatom. Phytoplankton group/species products were generated using the modified approach and validated with the reported events of red and green *Noctiluca* blooms in the Indian coastal waters. The present study also highlights two specific results based on MODIS retrieved time-series phytoplankton group/species image analysis: first, the observation of coexistence of diatom, red, and green *Noctiluca* in coastal waters of the north-western Bay of Bengal, and the second, phytoplankton community shift resulting in red/green *Noctiluca* proliferation following diatom.

Keywords Algal bloom · Diatom · Red *Noctiluca* · Green *Noctiluca* · Remote sensing · Bay of Bengal

S. K. Baliarsingh · R. M. Dwivedi (✉) · A. A. Lotliker · T. S. Kumar · S. S. C. Shenoi
Indian National Centre for Ocean Information Services, Hyderabad 500090, India
e-mail: rashmindwivedi@gmail.com

K. C. Sahu
Department of Marine Sciences, Behrampur University, Odisha 760007, India

Introduction

Phytoplankton blooms are the rapid increase in species cell density over time and such bloom events are of wide occurrence in the world ocean (Vinayachandran and Mathew 2003; Dwivedi et al. 2016; Baliarsingh et al. 2016). Sometimes, toxins released from certain phytoplankton species during bloom deteriorate ambient water quality and such events are categorized as "Harmful Algal Bloom (HAB)." The phytoplankton blooms are more often mono-specific, however sometimes caused by a mixture of species. In general, spatio-temporal variation in water quality parameters triggers phytoplankton growth leading to bloom (Anderson et al. 2002). Intense phytoplankton blooms have adverse impact on marine ecosystems and fishery resources (Naqvi et al. 1998). *Noctiluca scintillans* (hereafter *Noctiluca*), a common bloom forming dinoflagellate, occurs in two different forms, i.e., red and green (Harrison et al. 2011). The red *Noctiluca* exhibits heterotrophy mode of feeding by predating upon a wide array of phytoplankton species with utmost

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preference for diatom. The green *Noctiluca* develops endosymbiosis with a prasinophyte "*Pedinomonas noctilucae*" and accumulates energy from endosymbiont's photosynthesis (Madhu et al. 2012). However, the green *Noctiluca* also acts as a heterotroph upon good quantum of prey abundance (Zakaria and Ibrahim 2007; Gomes et al. 2014). Although *Noctiluca* is non-toxic, the blooms can result in fish mortality during termination phase due to depletion of dissolved oxygen in the water column and increase in concentration of ammonia. In general, *Noctiluca* blooms put significant influence on the food chain by voraciously predating upon phytoplankton species during heterotrophy as well as rapid utilization of nutrients during endosymbiotic autotrophy (Baliarsingh et al. 2016; Padmakumar et al. 2016).

Recurrent *Noctiluca* blooms have been observed in Indian Seas (Baliarsingh et al. 2016). Arabian Sea, the most productive component of the Indian Ocean, is experiencing annual episodes of green *Noctiluca* bloom in the northern part during winter (Gomes et al. 2014). In similar, events of red *Noctiluca* bloom are reported specifically in coastal waters of the Bay of Bengal (Baliarsingh et al. 2016).

Both forms of *Noctiluca* bloom deteriorate the ambient water quality and put adverse effect on aquatic ecology. Hence, regular spatio-temporal monitoring of *Noctiluca* blooms is essential to understand the preconditioning factor(s), onset, sustenance, termination, and environmental consequences. In this context, ocean color remote sensing provides scope of synoptic monitoring based on species specific spectral signatures of both forms of *Noctiluca* (Van Mol et al. 2007; Simon and Shanmugam 2012; Dwivedi et al. 2015). Earlier, a bio-optical algorithm was developed for remote detection of green *Noctiluca* using remote sensing reflectance (R_{rs}) of Moderate Imaging Spectroradiometer onboard Aqua satellite (MODIS) and found useful in studying the bloom dynamics in the Arabian Sea (Dwivedi et al. 2015, 2016). In comparison to green form of *Noctiluca*, the optical detection of red form needs careful assessment due to its heterotrophy mode of ecology. The intracellular content, specifically the diet, often determines the optical properties of red *Noctiluca* (Van Mol et al. 2007). Based on the above backdrop, the present study was carried out to tune the existing bio-optical algorithm enabling to detect red *Noctiluca* and discriminate from green *Noctiluca*.

Materials and methods

In situ bio-optic data collection and analysis

Time-series measurements of in situ biological, physico-chemical, and optical parameters were carried out in coastal waters north-western Bay of Bengal during a red *Noctiluca* bloom in April 2014 (Fig. 1). The details of physico-chemical-biological sample collection, analysis, and environmental consequences have been described in Baliarsingh et al. (2016).

The apparent optical properties were measured in the red *Noctiluca* bloom waters with aid of a free-falling optical profiler (Satlantic™ hyperspectal radiometer) at regular intervals from 1000 to 1400 h matching with equatorial crossing time of MODIS (1330 h). The upwelling/downwelling radiance/irradiance was recorded in spectral range of 350–800 nm. SatView™ software was used for data logging and Prosoft™ for raw data processing to different data levels. Subsequently, the spectral $R_{rs}(\lambda)$ was computed using the following equation

$$R_{rs}(0^+, \lambda) = \frac{L_w(0^+, \lambda)}{E_d(0^+, \lambda)}$$

where $L_w(0^+, \lambda)$ is the water leaving radiance (W m^{-2} nm^{-1} sr^{-1}) and $E_d(0^+, \lambda)$ is the downwelling irradiance (W m^{-2} nm^{-1}).

In order to discriminate the signal of red *Noctiluca* from green, in situ R_{rs} measured using hyperspectral radiometer (Satlantic™) during cruises of Indian research vessels (FORV *Sagar Sampada* cruise no. 314, 343 and ORV *Sagar Kanya* cruise no. 256) during green *Noctiluca* bloom events in the Arabian Sea were averaged to generate reference spectra. The reference spectra for red *Noctiluca* was generated by averaging the in situ R_{rs} measurements of red *Noctiluca* bloom event in coastal waters of the north-western Bay of Bengal.

Satellite data and processing

MODIS R_{rs} data available in different bands of visible spectrum (412 to 678 nm) were downloaded from NASA's OceanColor Web (http://oceancolor.gsfc.nasa.gov) for the dates corresponding to the reported occurrences of red *Noctiluca* bloom (Mohanty et al. 2007; Sahayak et al. 2005; Padmakumar et al. 2010;

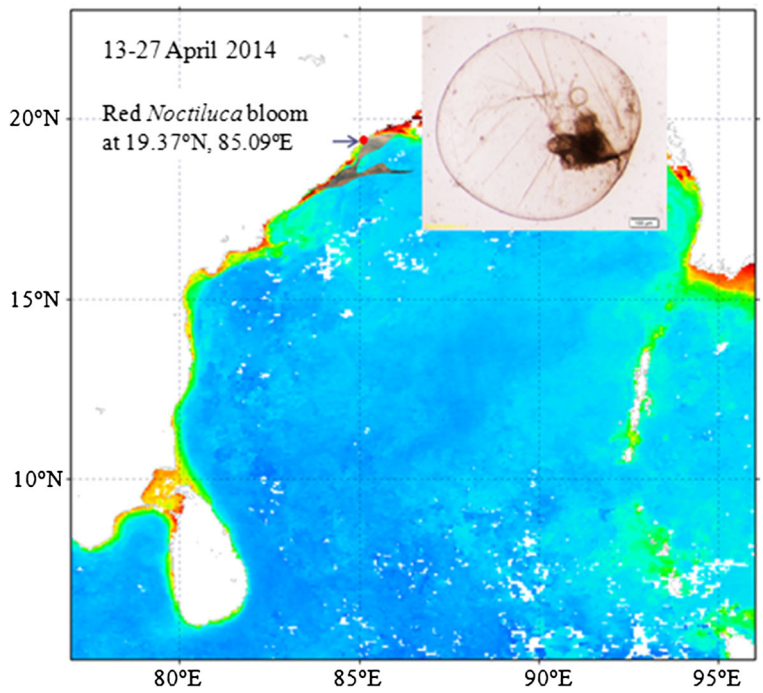


Fig. 1 MODIS retrieved chlorophyll image showing location of red *Noctiluca* bloom in coastal waters of the north-western Bay of Bengal

Gopakumar et al. 2009; Baliarsingh et al. 2016). Satellite retrieved sea surface temperature (SST), and chlorophyll products were downloaded from NASA's OceanColor web and processed in SeaDAS software. The R_{rs} data at different wavelengths were processed in image processing software ERDAS IMAGINE to generate phytoplankton class/species product. Detail of land/cloud masking and class allocation have been described in Dwivedi et al. (2015).

Results and discussion

Limitations of existing approach and background of the genesis of new approach

The earlier remote sensing approach developed by Dwivedi et al. (2015) for detection of green *Noctiluca* uses differential R_{rs} at 488 and 531 nm bands. The same approach was modified to detect red *Noctiluca* and to discriminate it from green *Noctiluca*. In general, red *Noctiluca* prominently backscatters in the red region of visible electromagnetic spectrum. Hence, MODIS R_{rs} at red wavelength bands 645, 667, and 678 nm were used in the present study to discriminate red *Noctiluca* from green. The lower magnitude of R_{rs} at red bands in case of green *Noctiluca* facilitated the remote discrimination of red and green *Noctiluca*. The approach of red *Noctiluca* detection was validated using the information on occurrence of this species in coastal waters of the north-western Bay of Bengal during April 2014 (Baliarsingh et al. 2016).

Technique for detection of red Noctiluca

Discrete in situ R_{rs} spectra measured in red *Noctiluca* bloom waters in coastal waters of the north-western Bay of Bengal during April were compared with corresponding MODIS bands (Fig. 2a). MODIS R_{rs}, which was derived using standard long wavelength atmospheric correction scheme, underestimated in situ R_{rs} (Fig. 2a). However, MODIS R_{rs} exhibited similarity in spectral shapes as like in situ. Phytoplankton group/species detection approach developed for green *Noctiluca* (Dwivedi et al. 2015) and the approach for red *Noctiluca* (present study) checks for the slopes (R_{rs} differences) of the segments from 488 to 531 nm and 667 to 678 nm as well as compares with a set of values determined from the archived data of different categories viz. red/green *Noctiluca*, diatom, non-bloom oceanic, and non-bloom coastal waters. R_{rs} differences were observed similar in both cases of in situ and satellite. In other words, the ranges of R_{rs} differences are wide enough to absorb the uncertainty in R_{rs} retrieval arising due to application of

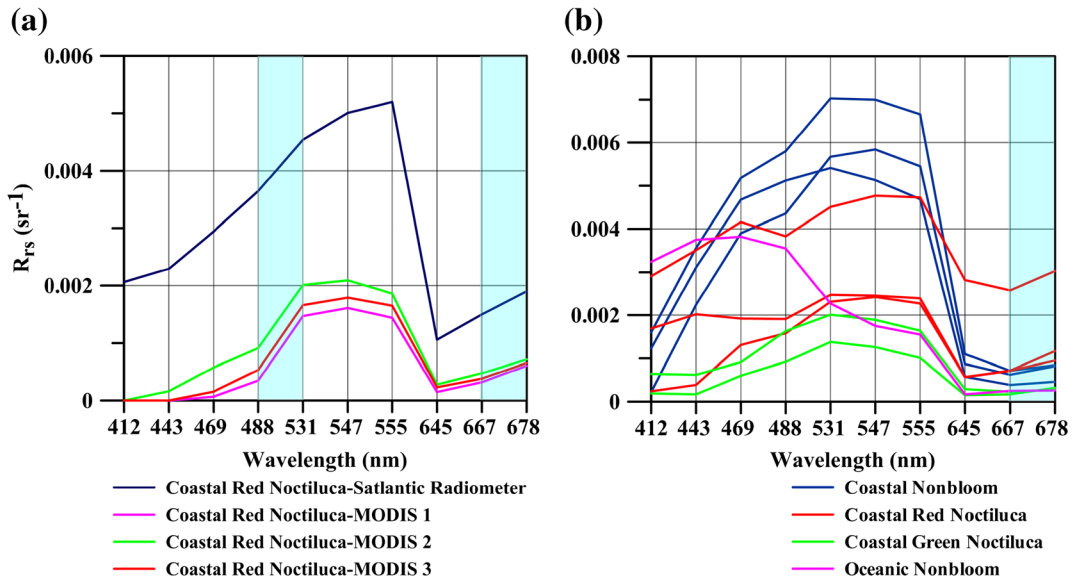


Fig. 2 **a** In situ remote sensing reflectance (R_{rs}) spectra of red *Noctiluca* measured using Satlantic hyperspectral radiometer. MODIS 1–3 represents MODIS retrieved R_{rs} for the reported red *Noctiluca* bloom events. **b** Typical R_{rs} spectra of red *Noctiluca*, green *Noctiluca*, coastal non-bloom, and oceanic non-bloom waters

standard atmospheric correction method in the coastal environment.

The absorption spectra of red *Noctiluca* exhibits peak at 488 nm and sharper decrease from 520 to 580 nm which causes relatively lower magnitude of R_{rs} at 488 nm relative to same at 531 nm (Astoreca et al. 2005; Van Mol et al. 2007). The same was observed from increasing trend of R_{rs} from 488 to 531 nm in case of *Noctiluca* (green as well as red) unlike the category of oceanic non-bloom waters (Fig. 2b). Although coastal non-bloom spectra also have increasing trend of R_{rs} from 488 to 531, these can be discriminated from *Noctiluca* using high slope value from 443 to 488 nm.

Subsequently, R_{rs} spectra for green *Noctiluca* was discriminated from red *Noctiluca* using 667 and 678 nm bands. Carotenoid (red pigment) is the dominant pigment in red *Noctiluca* (Balch and Haxo 1984), and hence, MODIS R_{rs} at red wavelength bands were used to discriminate red from green *Noctiluca*. There was a small decline in R_{rs} at 667 nm due to secondary absorption peak of chlorophyll-*a* and absorption by water. However, R_{rs} was significantly higher at 678 nm in the presence of red *Noctiluca* attributed to backscattering from red pigment. Thus, slope of the segment from 667 to 678 nm was relatively higher in case of red *Noctiluca*. Green *Noctiluca* exhibited almost flat response in the red bands (Fig. 2b).

Technique validation

Time-series in situ phytoplankton analyses confirmed bloom of red *Noctiluca* in coastal waters of the north-western Bay of Bengal (13–27 April 2014) (Baliarsingh et al. 2016). Phytoplankton group/species product was generated from MODIS Level-2 R_{rs} data using the approach developed in the present study for 16 April 2014 (one of the peak bloom days). The classified image confirmed the presence of red *Noctiluca* at the same location (Fig. 3a). In another comparison, phytoplankton product generated from MODIS for 5 April 2005 also matched with bloom event of red *Noctiluca* at the same location (Fig. 3b) (Mohanty et al. 2007).

Physical preconditioning for red Noctiluca proliferation

The red *Noctiluca* bloom was detected in coastal waters of the north-western Bay of Bengal during 13–27 April 2014 (Baliarsingh et al. 2016). In general, East India Coastal Current (EICC) flows north-eastward in the Bay of Bengal from February till September with a strong peak in March–April (Schott and McCreary 2001). Due to Coriolis effect, high magnitude EICC in April stimulates Ekman transport of the water mass resulting upwelling and subsequent nutrient entrainment from subsurface to surface. Signature of lower SST due to coastal

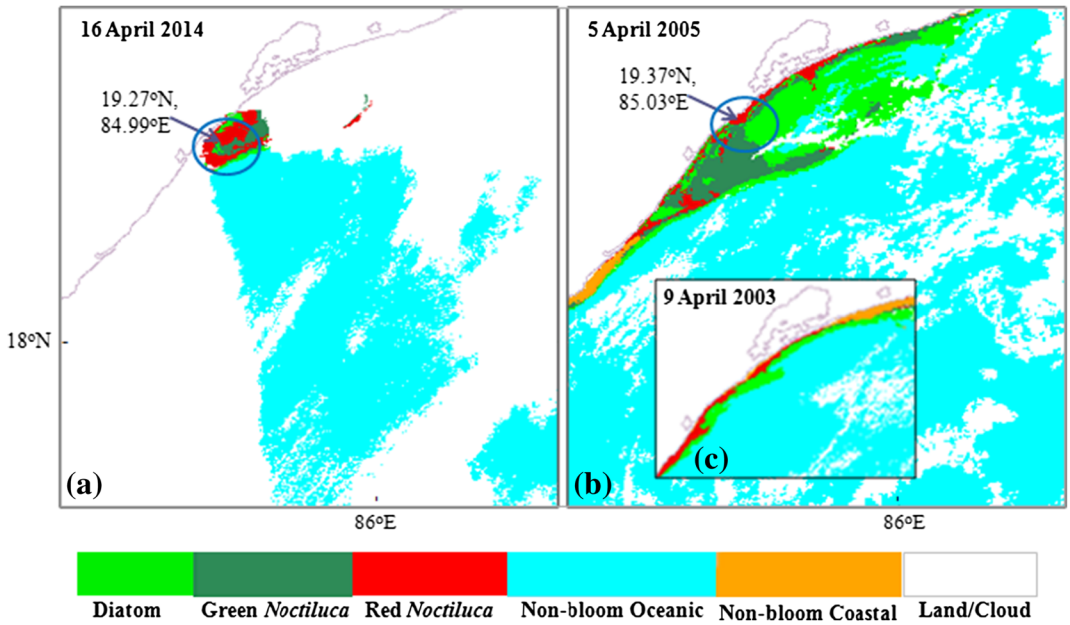


Fig. 3 Phytoplankton group/species product generated from MODIS **a** 16 April 2014, **b** 5 April 2005, and **c** 9 April 2003 in coastal waters of the north-western Bay of Bengal

upwelling at 19°N and resultant higher magnitude of chlorophyll-*a* (proxy of phytoplankton biomass) was observed from satellite images of 5 April 2005 (Fig. 4). Availability of inorganic macronutrients, especially silicate, fuels to diatom growth. In general, the red as well as green *Noctiluca* feed voraciously on phytoplankton species with a special preference for diatoms (Buskey 1995; Kiφrboe and Titelman 1998).

Role of water temperature in red Noctiluca outbreak

In general, diatoms are the preferred prey for red *Noctiluca*. Hence, diatom availability is considered as an important conducive factor especially for red *Noctiluca* growth. However, high density of diatom is not always a sufficient condition to precondition the ambient medium for red *Noctiluca* outbreak. Apart from

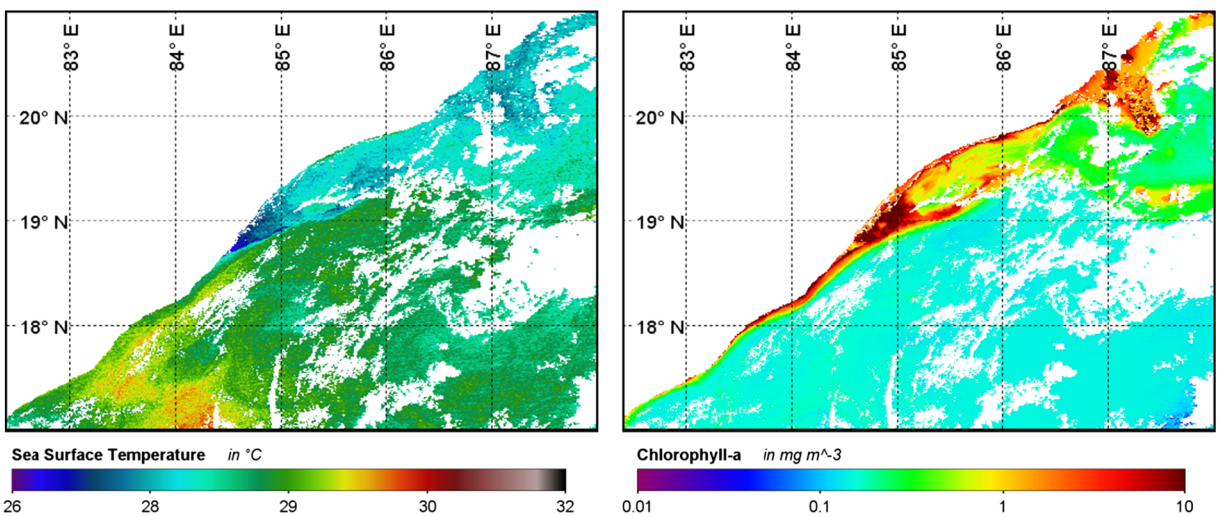


Fig. 4 MODIS retrieved sea surface temperature (SST) (*left panel*) and chlorophyll (*right panel*) images corresponding to red *Noctiluca* bloom in coastal waters of north-western Bay of Bengal

the prey availability, water temperature plays a pivotal role in conditioning the environment suitable for red *Noctiluca* growth. Relatively low water temperature was reported to act as a conducive condition for red *Noctiluca* bloom (Sahayak et al. 2005). Relatively lower magnitude of SST (28.5 °C) was observed during the red *Noctiluca* bloom period in April 2005 in the present study site (Mohanty et al. 2007). Concomitantly, the recent red *Noctiluca* bloom observed in the same site in the Bay of Bengal during April 2014 was also associated with similar magnitude of SST during peak phase. Further, the bloom crashed with increase in SST (Baliarsingh et al. 2016). Hence, phytoplankton class products depicting red *Noctiluca* bloom were correlated with SST for different bloom events in the coastal waters off Gopalpur, north-western Bay of Bengal in order to decipher the role of water temperature. The following case studies explain the SST preference of red *Noctiluca*.

April 2014: SST was in a range 26–29 °C during the pre-bloom period (March) and the initial stage of *Noctiluca* proliferation (in early April). A prominent patch of red *Noctiluca* corresponded to SST of 28.8 °C on 16 April 2014 (Fig. 3a). SST varied from 27 to 31 °C during the mid to late bloom phase (late April) (Baliarsingh et al. 2016).

April 2005: Relatively lower water temperature (28.5 °C) prevailed on 5 April 2005 during initial outbreak of red *Noctiluca* bloom. In agreement, the signature of red *Noctiluca* was also observed from satellite imagery (Fig. 3b). In corroboration, suitability of low water temperature for red *Noctiluca* proliferation was also reported for this bloom event (Mohanty et al. 2007).

April 2003: A bloom of red *Noctiluca* was observed from time-series satellite data analysis on 9 April 2003. Relatively warmer SST (> 29 °C) was observed from MODIS data in far coastal waters. The red *Noctiluca* was observed only with thin spread very close to the coast associated with relatively cooler SST. The red *Noctiluca* did not develop in a larger patch in spite of diatom availability could be due to non-suitability of temperature (Fig. 3c).

Elbrachter and Qi (1998) have mentioned that temperature affects mean doubling time of *Noctiluca* cells. Thus, although coastal upwelling during summer in the Bay of Bengal is a recurring process, red *Noctiluca* does not occur every year or occurs in small patches for short duration that remains un-noticeable.

Coexistence of red and green Noctiluca with respect to space and time

Red and green *Noctiluca* can occur in the same location but at different times (Harrison et al. 2011). Green *Noctiluca* occurs in the high productive upwelling waters while red *Noctiluca* develops in oligotrophic warmer waters (Devassy 1989; Harrison et al. 2011). However, MODIS retrieved phytoplankton products of 16 April 2014 and 5 April 2005 that revealed presence of green *Noctiluca* in vicinity of red *Noctiluca* bloom regions (Fig. 3a, b). This observation indicated the coexistence of red and green *Noctiluca* in the same space and time. The following discussion substantiates this detection.

Gopakumar et al. (2009) have reported green *Noctiluca* proliferation in the coastal waters of Gulf of Mannar (south-east coast of India) during 2–10 October 2008. In an another report, Eashwar et al. (2001) have observed green *Noctiluca* bloom off Port Blair, the Bay of Bengal in July 2000 which turned the coastal waters to bright green color. On the west coast of India, Sulochanan et al. (2014) have observed green *Noctiluca* bloom in the coastal waters of Mangalore on 12 May 2011. Concisely, green *Noctiluca* does occur in the coastal waters of both the coasts of India. MODIS data were processed for detection of green *Noctiluca* reported by Gopakumar et al. (2009) and confirmed the presence of green *Noctiluca* along southeast coast of India (Fig. 5). In addition to the satellite-based confirmation on green *Noctiluca* along Indian coastal waters, the technique of detecting green *Noctiluca* (Dwivedi et al. 2015) proved to perform well in the coastal waters of Bay of Bengal.

Green as well as red *Noctiluca* is associated with high productivity attributed to convection/upwelling. Both the forms of *Noctiluca* are voracious predators and require abundant food supply, mainly diatoms in order to maintain high growth rate (Schaumann et al. 1988; Nakamura 1998; Elbrachter and Qi 1998). The extensive spread of diatom over the neighboring area of green as well as red *Noctiluca* was observed from MODIS phytoplankton product of 5 April 2005 (Fig. 3b).

In general, water temperature plays an important role in controlling the distribution and abundance of *Noctiluca*. However, it is not a significant conflicting

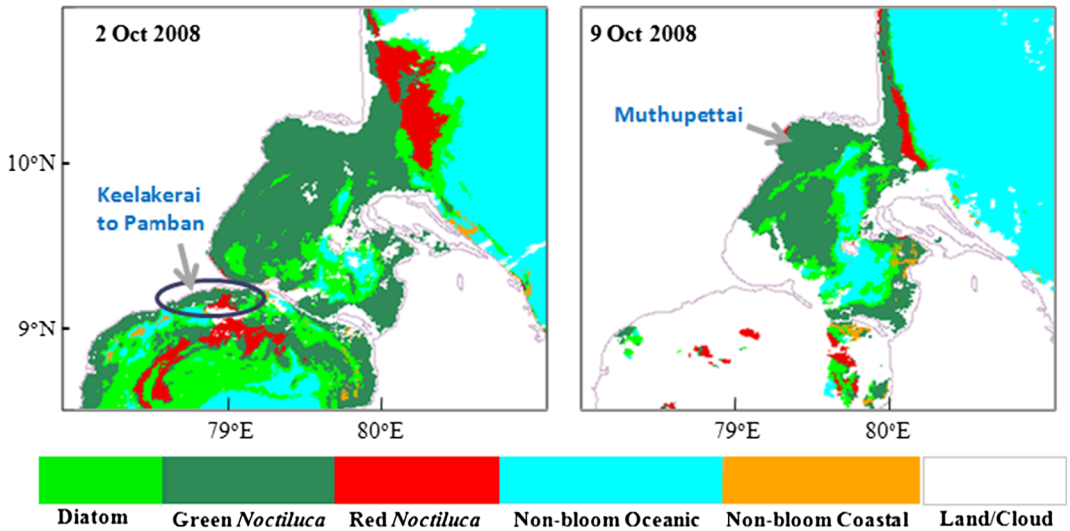


Fig. 5 MODIS retrieved phytoplankton group/species image corresponding to green *Noctiluca* bloom event in southeast coast of India

factor for coexistence of both forms of *Noctiluca*. Harrison et al. (2011) have documented a water temperature range of 25–30 °C for green *Noctiluca*. Gopakumar et al. (2009) have mentioned 29.5 °C temperature during waning phase of green *Noctiluca* bloom in south-east coast of India. Sulochanan et al. (2014) have reported water temperature in the range of 28.2–33 °C during green *Noctiluca* proliferation in the west coast of India. In context of red *Noctiluca*, Harrison et al. (2011) have reported a water temperature range of 29–30 °C for Indian waters. Mohanty et al. (2007) have observed red *Noctiluca* in the coastal waters of north-western Bay of Bengal at water temperature of 28.5 °C. At the same location, red *Noctiluca* was observed on 16 April 2014 with water temperature of 28.8 °C. The above observations confirmed no conflict in water temperature preference between two forms of *Noctiluca*. Thus, it is apparent that both forms of *Noctiluca* can occur at the same time attributed to favorable temperature and prey availability. SST was studied for green as well as red *Noctiluca* pixels of 5 April 2005 and observed in the range of 28–29 °C irrespective of species forms (Fig. 4).

Reference spectra were generated for red and green *Noctiluca* by averaging R_{rs} spectra of the bloom events (Fig. 6). On this, R_{rs} spectra from red and green *Noctiluca* pixels of 5 April 2005 were superimposed. Although magnitudes of R_{rs} were different from reference spectra, similar spectral shapes were observed. In other words, range of the slopes for the segments of 488–531 and 667–678 nm coincided with the reference spectra. With the above verification, the satellite-based detection of red and green *Noctiluca* co-occurrence at the same time in vicinity is confirmed. While in situ observations are restricted to finite distances, synoptic coverage of satellite retrieved phytoplankton group/species classified products facilitates the simultaneous detection and monitoring of both red and green *Noctiluca*. Hence, it is hypothesized that when the prey (diatom) is consumed locally, red *Noctiluca* might be developing endosymbiosis and would appear green.

Case study on determining the role of prey and temperature in proliferation and coexistence of red and green Noctiluca

Noctiluca blooms are known to occur subsequent to diatom proliferation (Kiorboe 1998; Tiselius and Kiorboe 1998; Dela-Cruz et al. 2002). Diatom exhibits eurythermal ecology and can proliferate under abundant nutrient supply subsequent to upwelling or convection, even in the turbulent waters, whereas *Noctilua* prefers stable and specific water temperature (Dwivedi et al. 2016). In order to study this environmental phenomenon, the bloom scenario of March 2005 in coastal waters of the north-western Bay of Bengal was studied through MODIS retrieved SST and phytoplankton group/species products. Temporal variability in SST revealed signatures of upwelling during 22–29 March 2005 (Fig. 7a, b). By the time upwelling ceased and water started warming, diatom was fully evolved (marked as ellipse in Fig. 7d). Diatom is seen replaced by red and green *Noctiluca* in 5

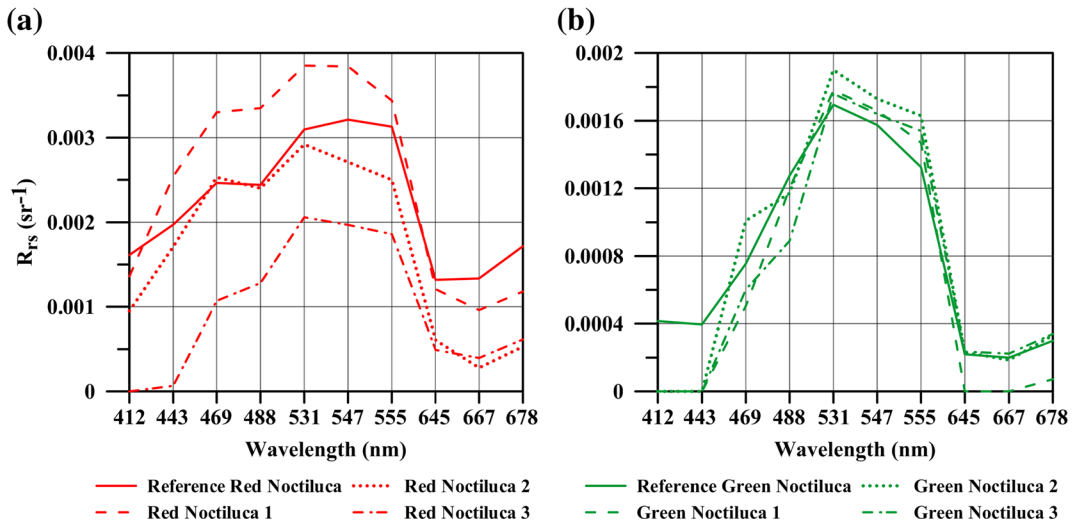


Fig. 6 **a** Remote sensing reflectance (R_{rs}) of red *Noctiluca*. Reference spectra represents average of in situ radiometric measurements in coastal waters of north-western Bay of Bengal during red *Noctiluca* bloom event in April, 2014. MODIS retrieved R_{rs} spectra for 5 April 2005 in coastal waters of north-western Bay of Bengal are presented as red *Noctiluca* 1–3. **b** R_{rs} of green *Noctiluca* bloom. Reference spectra represents average of in situ radiometric measurements in the Arabian Sea during green *Noctiluca* bloom. MODIS retrieved R_{rs} spectra for 5 April 2005 off Gopalpur (in vicinity of red *Noctiluca* proliferated area) are presented as green *Noctiluca* 1–3

April 2005 image, after a period of 2 weeks. It is believed that availability of favored prey (diatom) and suitable water temperature resulted in *Noctiluca* proliferation. Time-series analysis of phytoplankton group/species products demonstrated shift in community dominance from diatom to red/green *Noctiluca*.

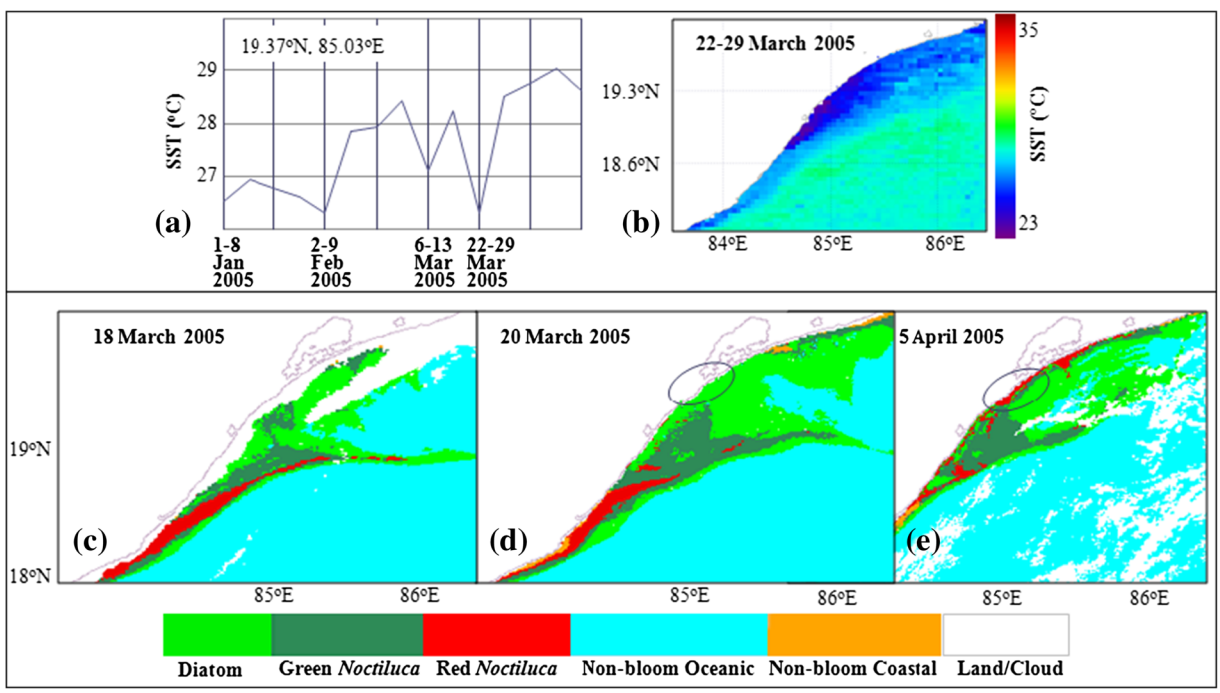


Fig. 7 **a** MODIS retrieved 8-day rolled sea surface temperature (SST). **b** MODIS retrieved spatial map of SST. **c**–**e** Phytoplankton group/species image generated from MODIS of coastal waters of the north-western Bay of Bengal

Conclusion

In summary, based on the principle of specific spectral shapes of R_{rs} of different phytoplankton classes/species, the existing technique of detecting green *Noctiluca* using MODIS R_{rs} was appropriately modified to detect bloom of red *Noctiluca* in the coastal waters of the north-western Bay of Bengal. Addition of MODIS R_{rs} at longer wavelengths viz. 667 and 678 nm in the existing algorithm performed well in detecting both green and red forms of *Noctiluca*. Conducive environmental conditioning of coastal waters resulted phytoplankton community shift from diatom to *Noctiluca*. The spatio-temporal coexistence of red and green *Noctiluca* observed in coastal waters of the north-western Bay of Bengal could be attributed to prey availability and conducive temperature. The outcomes of the present study can be utilized to monitor *Noctiluca* blooms in Indian coastal water based on satellite retrieved indicators such as (i) preponderance of diatom and (ii) conducive SST.

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